

Speech Signal Processing - Module 2

Topic: Short-Time Fourier Transform (STFT) and Linear Prediction Analysis (LPA)

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1. Introduction

Speech is a non-stationary signal, meaning its characteristics change over time. However, within very short intervals (typically 20–30 ms), it can be considered quasi-stationary. This assumption forms the basis for most speech analysis techniques. During these short windows, the vocal tract configuration remains relatively constant, allowing for localized time-frequency analysis using the Short-Time Fourier Transform (STFT) or parametric modeling using Linear Prediction Analysis (LPA).

2. Short-Time Fourier Transform (STFT)

The Fourier Transform provides frequency-domain information but assumes stationarity, which is not valid for speech. The STFT addresses this by applying the Fourier Transform to small overlapping segments of the speech signal.

Mathematically, STFT is expressed as:

$$\text{STFT}\{x[n]\}(m, k) = \sum x[n + m] \cdot w[n] \cdot e^{-j2\pi kn/N}$$

where $w[n]$ is the window function, m is the time index (frame position), and k is the frequency bin.

The STFT output is a time–frequency representation showing how the spectral content of the signal evolves over time. This is typically visualized as a spectrogram.

Advantages:

- Captures time-varying frequency information.
- Basis for perceptual features like Mel-Frequency Cepstral Coefficients (MFCCs).
- Mimics the human auditory system's cochlear analysis through Mel and Gammatone filterbanks.

Limitations:

- Trade-off between time and frequency resolution (Heisenberg's uncertainty principle).
- Choice of window size affects analysis accuracy.

3. Linear Prediction Analysis (LPA)

Linear Prediction Analysis models the speech signal as a linear combination of its past samples. It assumes the speech signal is generated by passing an excitation source

(voiced/unvoiced) through a vocal tract filter represented by an all-pole model.

Mathematical formulation:

$$s(n) = \sum a_k * s(n - k) + e(n), \quad k = 1 \text{ to } p$$

where 'a_k' are predictor coefficients, 'e(n)' is the prediction error (residual), and 'p' is the prediction order.

The residual represents the excitation signal, while the filter coefficients represent the vocal tract response.

Transfer Function:

$$H(z) = 1 / (1 - \sum a_k * z^{-k})$$

Applications:

- Formant estimation (vocal tract resonances).
- Pitch detection (via LP residual).
- Speech coding (LPC coders).
- Speech synthesis and recognition.

Advantages:

- Compact parametric model of the speech spectrum.
- Aligns with the source-filter model of speech.

Limitations:

- Assumes an all-pole model, which may not fit nasal or fricative sounds accurately.
- Model order selection critically affects performance.

4. STFT vs LPA – A Comparative View

STFT and LPA both operate under the short-time stationarity assumption but differ fundamentally in approach:

STFT:

- Non-parametric (uses windowed DFT).
- Provides full spectral information.
- Used for perceptual and frequency-based analysis.

LPA:

- Parametric (models spectral envelope via poles).
- More compact representation.
- Provides interpretable parameters (formants, residual).

Together, these methods form the foundation of modern speech analysis, feature extraction, and synthesis.

5. Summary

- Speech is quasi-stationary in short intervals.
- STFT offers time–frequency analysis; LPA provides parametric modeling.
- STFT underpins perceptual feature extraction methods like MFCCs.
- LPA models vocal-tract characteristics and residual excitation.
- Both techniques are crucial in speech recognition, synthesis, and coding.